Volume III.

§ 97. Orders and ideals.

Let $\mathfrak{m}$ be an ideal of the quadratic field Ω. If all whole numbers divisible by $\mathfrak{a}$ are divisible by a rational number m, then the ideal $\mathfrak{a} = m\mathfrak{m}$ is divisible by m; one has

$$N(\mathfrak{a}) = m^2 N(\mathfrak{m}),$$

and the rational number $mN(\mathfrak{m})$, which is smaller than $N(\mathfrak{a})$, is divisible by $\mathfrak{a}$.

If m is the greatest natural number by which $\mathfrak{a}$ is divisible, we shall call m the *divisor* of $\mathfrak{a}$. If the divisor is $= 1$, the ideal is called *primary*; otherwise *derived.* One obtains all ideals derived from a primary ideal $\mathfrak{m}$ by multiplying $\mathfrak{m}$ by arbitrary natural numbers, and all ideals derived from one primary ideal are mutually equivalent.

3. *Theorem.* If $\mathfrak{a}$ is a primary ideal, then the norm of $\mathfrak{a}$ is at the same time the smallest natural number divisible by $\mathfrak{a}$.

For let the smallest natural number divisible by $\mathfrak{a}$ be denoted by a; then the norm of $\mathfrak{a}$ is in any case divisible by a. We put

$$N(\mathfrak{a}) = ma,$$

and, if $\mathfrak{a}'$ is the ideal conjugate to $\mathfrak{a}$, it follows, since by Vol. II, §164 the norm of an ideal is equal to the product of the conjugate ideals, that

$$N(\mathfrak{a}) = \mathfrak{a}\mathfrak{a}' = ma. \tag{1}$$

Since a is to be divisible by $\mathfrak{a}$, we put

$$a = \mathfrak{a}\mathfrak{m}' = \mathfrak{a}'\mathfrak{m}$$

and obtain from (1):

$$\mathfrak{a} = m\mathfrak{m}. \tag{2}$$

Thus m enters the divisor of the ideal $\mathfrak{a}$, and thereby Theorem 3 is proved, since, when $\mathfrak{a}$ is primary, its divisor must be $= 1$.

4. *Theorem.* If $\mathfrak{m}$ is a primary ideal of the field Ω, then every whole number in Ω is congruent modulo $\mathfrak{m}$ to a rational number.

For, by Vol. II, §165, $N(\mathfrak{m})$ is in general the number of numbers incongruent modulo $\mathfrak{m}$ in Ω. If now $N(\mathfrak{m}) = a$ is at the same time the smallest natural number divisible by $\mathfrak{m}$, then the a numbers

$$0, 1, 2, \ldots, a-1$$

are all incongruent, and among them all number classes modulo $\mathfrak{m}$ are represented.

Let now $[Q]$ be an order with conductor Q and discriminant D, and let $(1, \theta)$ be the basis of this order. Further, let $\mathfrak{m}$ be a primary ideal prime to Q. By Theorem 4 there is then a whole rational number t satisfying the condition

$$\theta + t \equiv 0 \pmod{\mathfrak{m}}.$$

We put

$$\begin{aligned} t &= \frac{b}{2}, && \text{if } D \equiv 0 \pmod 4, \\ t &= \frac{1+b}{2}, && \text{if } D \equiv 1 \pmod 4, \end{aligned} \tag{3}$$

and obtain, by §96, (5),

$$\theta + t = \frac{b + \sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{m}}, \tag{4}$$

where b is a whole rational number satisfying, by (3), the condition

$$b \equiv D \pmod{2}.$$

In addition let

$$N(\mathfrak{m}) = a, \tag{5}$$

so that a is the smallest natural number divisible by $\mathfrak{m}$.

By the congruence (4), the number b is determined only modulo $2a$. For if b, b' are two numbers which satisfy the condition (4), then $\frac{1}{2}(b' - b)$ is divisible by $\mathfrak{m}$.

Further, since

$$N\left(\frac{b+\sqrt{D}}{2}\right) = \frac{b^2 - D}{4}$$

is a whole rational number divisible by $\mathfrak{m}$, it must be divisible by a; if we put it equal to ac, then c is a whole rational number and

$$D = b^2 - 4ac, \tag{6}$$
$$b^2 \equiv D \pmod{4a}, \tag{7}$$

and, if the congruence (7) is fulfilled, then every number b' congruent to b modulo $2a$ satisfies the same congruence.

We infer from this:

5. *Theorem.* Every primary ideal $\mathfrak{m}$ prime to Q, whose norm is $= a$, gives us by (4) one and only one root b of the congruence (7), if by roots one understands not individual numbers but number classes modulo $2a$.

The two numbers

$$a, \qquad \frac{b+\sqrt{D}}{2} \tag{8}$$

if a is relatively prime to Q, cannot both be divisible by one and the same rational prime number; for otherwise

$$\frac{b+\sqrt{D}}{2} - \frac{b-\sqrt{D}}{2} = \sqrt{D}$$

would have to be divisible by p; consequently D would be divisible by p^2, and p would have to be a divisor of Q, contrary to the assumption.

For an odd p this is evident; for $p = 2$, however, it would follow that

$$b \equiv 0 \pmod{2}, \qquad b^2 - D \equiv 0 \pmod{16},$$

and hence

$$D \equiv 0, 4 \pmod{16}.$$

Thus $D/4$ would still be a discriminant and 2 would have to enter Q. This proves:

6. *Theorem.* If a is a natural number prime to Q and b is a root of (7), then one obtains a definite primary ideal $\mathfrak{m}$ as the greatest common divisor of the two numbers (8).

We now prove:

7. *Theorem.* Every number μ of the order $[Q]$, divisible by the primary ideal $\mathfrak{m}$ and prime to Q, is representable once and only once in the form

$$\mu = ax + \frac{b+\sqrt{D}}{2}y, \tag{9}$$

where x, y are whole rational numbers. If x, y are regarded as variables, then μ is called a basis form of the ideal $\mathfrak{m}$ in the order $[Q]$.

For the proof we note that every number of the order $[Q]$, by (4), is contained in the form

$$\omega = z + \frac{b+\sqrt{D}}{2}y, \tag{10}$$

where y and z are whole rational numbers. If now $\mathfrak{m}$ is primary, then

$$N(\mathfrak{m}) = a \tag{11}$$

is the smallest natural number divisible by $\mathfrak{m}$, and therefore, if ω is to be divisible by $\mathfrak{m}$, z must be divisible by a. Thus, putting $z = ax$, we obtain the form (9). Conversely, it is evident that every number of this form is divisible by $\mathfrak{m}$.

That the representation of a number μ in this form is possible in only one way follows from the fact that $\sqrt{D}$ is irrational.

The form μ is therefore called a basis form of the ideal $\mathfrak{m}$ in the order Q. From it one obtains a basis form of the same ideal in the field Ω:

$$\lambda = ax + \frac{b'+\sqrt{\Delta}}{2}y,$$

where b' is determined from the congruence

$$b'Q \equiv b \pmod{2a},$$

which is always solvable, since Q is relatively prime to a, and, for even Q, b too is even.

From (9) one obtains

$$N(\mu) = a(ax^2 + bxy + cy^2). \tag{12}$$

The number μ obtained by assigning definite numbers to x, y in (9) can be decomposed into ideal factors:

$$\mu = \mathfrak{a}\mathfrak{m},$$

if

$$N(\mathfrak{a}) = ax^2 + bxy + cy^2, \tag{13}$$

and the following theorem holds:

8. *Theorem.* If μ is relatively prime to Q, and x, y have no common divisor, then the ideal $\mathfrak{a}$ is primary.

We prove it thus: let p be a rational prime number entering $\mathfrak{a}$; then μ is divisible by p, and from this it follows first that y must be divisible by p; for a number ω of the form (10) can be divisible by a prime number p not entering Q only when both y and $z = ax$ are divisible by p. Therefore, since by assumption x and y are relatively prime, x cannot be divisible by p, and a must be divisible by p.

Put $a = p^k a_1$, where a_1 is no longer divisible by p, and denote by $\mathfrak{p}$ a prime ideal entering p. Then, by (11), $\mathfrak{p}$ must enter $\mathfrak{m}$ or $\mathfrak{m}'$, and, since $\mathfrak{m}$ has been assumed primary, $\mathfrak{p}$ cannot be $= \mathfrak{p}'$. Hence $N(\mathfrak{p}) = p = \mathfrak{p}\mathfrak{p}'$, and if $\mathfrak{m}$ is divisible by $\mathfrak{p}$, it cannot at the same time be divisible by $\mathfrak{p}'$. Consequently $\mathfrak{m}$ is divisible by $\mathfrak{p}^k$. Thus, by the definition of $\mathfrak{m}$ (Theorem 6),

$$\frac{b+\sqrt{D}}{2}$$

is also divisible by $\mathfrak{p}^k$, and consequently

$$\frac{b+\sqrt{D}}{2}y$$

is divisible at least by $\mathfrak{p}^{k+1}$, while ax, and hence μ, is divisible exactly by $\mathfrak{p}^k$. On the other hand $\mathfrak{a}\mathfrak{m} = \mu$ is likewise divisible at least by $\mathfrak{p}^{k+1}$, which is a contradiction. The same holds also when

$p = \mathfrak{p}^2$, that is, when p enters Δ but not Q. Consequently no rational prime number can be contained in $\mathfrak{a}$, and $\mathfrak{a}$ is primary.

From what has so far been proved we can formulate the following theorem:

9. *Theorem.* If

$$\mu = ax + \frac{b+\sqrt{D}}{2}y = \mathfrak{m}\mathfrak{a},$$

$$\mu' = ax' + \frac{b+\sqrt{D}}{2}y' = \mathfrak{m}\mathfrak{a}'$$

are two numbers prime to Q, neither x and y nor x' and y' have a common divisor,

$$N(\mathfrak{a}) = N(\mathfrak{a}') = A,$$

and, for one and the same number B,

$$\frac{B+\sqrt{D}}{2} \equiv 0 \pmod{\mathfrak{a}}, \qquad \equiv 0 \pmod{\mathfrak{a}'},$$

then the two ideals $\mathfrak{a}$ and $\mathfrak{a}'$ are identical, and the numbers μ, μ' differ only by a unit factor.

For both $\mathfrak{a}$ and $\mathfrak{a}'$ are, by 8., primary, and are, by 6., defined as greatest common divisors of

$$A \quad \text{and} \quad \frac{B+\sqrt{D}}{2}.$$